

NLP Research Project — Qaybinta Shaqada (4 Qof)

QOF 1 — Data & Preprocessing Specialist

STEP 1 — Data Collection

- Soo ururi dataset-ka
- Hubi text column iyo label column
- Ka saar duplicate rows iyo empty rows
- Dataset kala saar: 70% Train, 15% Validation, 15% Test
- Kaydi train.csv, val.csv, test.csv

STEP 2 — Annotation & Label Verification

- Dib u eeg labels-ka
- Sax khaladaadka
- Isticmaal hal annotation format
- Samee inter-annotator agreement
- Kaydi final labels

STEP 3 — Preprocessing

- Lowercasing
- Remove duplicates
- Remove unnecessary symbols
- Text normalization
- Tokenization
- Kaydi clean_train.csv, clean_val.csv, clean_test.csv

STEP 7 — Tokenization (Hugging Face)

- Install transformers
- Implement tokenizer
- Tijaabi XLM-R tokenizer
- Convert input_ids iyo attention_masks
- Document tokenizer settings

QOF 2 — EDA & Traditional ML Specialist

STEP 4 — Exploratory Data Analysis (EDA)

- Class distribution chart
- Word frequency analysis
- Most common tokens
- Sentence length analysis
- Visualization charts
- Isticmaal matplotlib iyo seaborn

STEP 5 — Traditional ML Models

- TF-IDF feature engineering
- Word n-grams iyo character n-grams
- Train Logistic Regression
- Train Linear SVM
- Train Random Forest
- Train XGBoost
- GridSearchCV hyperparameter tuning
- Evaluate using Accuracy, Precision, Recall, F1-score

QOF 3 — Deep Learning & Transformer Specialist

STEP 6 — Deep Learning Models

- Build BiLSTM architecture
- Isticmaal Word2Vec iyo FastText embeddings
- Train and validate models
- Prevent overfitting using Dropout iyo Early Stopping
- Save checkpoints iyo training logs

STEP 8 — Transformer Fine-tuning

- Fine-tune mBERT
- Fine-tune XLM-RoBERTa
- Fine-tune SomBERTa, AfroXLMR, AfriBERTa
- Save best checkpoint
- Generate accuracy comparison table

QOF 4 — Evaluation, XAI & Research Writing Specialist

STEP 9 — Evaluation

- Evaluate all models
- Calculate Accuracy, Precision, Recall, F1-score
- Generate confusion matrix

STEP 10 — Explainable AI (XAI)

- Implement SHAP
- Implement LIME
- Feature importance analysis
- Error analysis

STEP 11 — Model Comparison

- Compare all ML, DL, iyo Transformer models
- Identify best model
- Write strengths iyo weaknesses

STEP 12 — Research Discussion & Final Paper

- Write findings
- Discuss challenges
- Add future work
- Write conclusion
- Prepare final research paper